

ASPERITAS

CODEX AI AGENT BUILD GUIDEBOOK

Biological Intelligence Factory / Source-Grounded Agent Construction Manual

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Purpose

이 PDF 는 새 채팅 또는 Codex 세션에 업로드/복붙하여, 자료 수집 단계부터 자체 AI 에이전트 구축까지 단계별로 실행하기 위한 총괄 가이드북이다.

핵심 원칙: 먼저 source-grounded internal decision system 을 만든다. 완전 자율 실험실 에이전트는 후순위다.

주의: 본 문서는 접근 가능한 업로드 자료, AOS v8.0/v9.0/v10.0, 이전 대화에서 정리된 전략/소스맵을 통합한 구축 명세서다. 실제 production vector DB, KG, eval suite, 법적 라이선스 검토, 전체 웹 크롤링, wet-lab validation 완료를 의미하지 않는다.

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0. Executive Bottom Line

Bottom Line

Asperitas 가 지금 만들어야 하는 것은 “멋진 챗봇”이 아니라, 내부 자료와 외부 검증 소스를 분리하고, 모든 답변을 출처 기반으로 만들며, 전략/연구/컴플라이언스/운영 판단을 반복 개선하는 Source-Grounded Biological Intelligence System 이다. Codex 는 이 시스템의 코드/폴더/파일프라인을 만드는 실행 엔진으로 써야 한다.

이 가이드북의 최종 산출물은 다음 네 가지다.

- 1) Asperitas 의 모든 문서를 안전하게 분류/수집/처리하는 Knowledge Base 구조
- 2) AOS v10.0 을 실행 규칙으로 삼는 RAG + Agent Harness
- 3) COO/전략/컴플라이언스/DBTL/시장/그랜트 업무를 처리하는 MVP agent 세트
- 4) 출처/평가/로그/승인/보안 체계를 갖춘 내부 의사결정 인프라

가장 중요한 경계선: 이 프로젝트의 1 단계 목표는 완전 자율 생물 실험 에이전트가 아니다. 먼저 사내 문서, 전략, 규제, 연구자료를 정확히 읽고 출처를 달아 판단을 보조하는 내부 결정 시스템을 만든다. 그 다음 실험 데이터 레지스트리, 프로토콜 버전관리, 샘플 추적, 이미지/표현형 파이프라인, LIMS/ELN-ready 구조로 확장한다.

Decision	Hard Rule
Start with RAG + source registry	No agent without source-grounding, metadata, and evals.
Do not claim “fully learned”	Only say production DB exists after ingestion/license/chunking/embedding/vector DB/KG/eval deployment are verified.
Human approval first	All regulatory, biosafety, investor, legal, wet-lab, or IP-sensitive outputs require review.
Build the data layer before foundation model language	A foundation model claim requires proprietary, validated, diverse, metadata-rich data and repeatable DBTL loops.
Compliance is infrastructure	CITES/Nagoya/LMO/biosafety/data security cannot be patched later.

1. 사용법: 새 채팅/Codex 에 이 PDF 를 넣고 시작하는 법

새 채팅에서 이 PDF 를 쓰는 방식은 두 단계다. 첫째, 이 PDF 를 업로드하고 “이 문서를 기준으로 Asperitas agent repository 를 단계별로 구축하라”고 지시한다. 둘째, 실제 원본 자료는 별도 파일로 업로드하거나 아래 폴더 구조에 맞춰 배치한다. 이 PDF 만으로 원문 전체가 자동 수집 되는 것은 아니다.

1.1 새 채팅 첫 메시지

너는 Asperitas 내부 AI agent 구축 담당 Codex/AI engineer 다.
첨부한 “Asperitas Codex AI Agent Build Guidebook v1.0” 을 최상위 실행 문서로 삼아라.
목표는 자료 수집 -> source registry -> metadata -> redaction/license review -> markdown extraction -> chunking -> embeddings/vector DB -> RAG -> agent harness -> eval suite -> compliance guardrails -> internal MVP app 구축이다.
먼저 이 PDF 를 요약하지 말고, 실행 가능한 repository 구조와 파일 생성 순서부터 제시하라.
중요: 실제 production DB 가 아직 구축되지 않았다는 truth boundary 를 지켜라. 모든 산출물은 source-grounded, human-approved, audit-ready 구조로 만들어라.

1.2 자료 업로드 순서

21. AOS v10.0, v9.0, v8.0 을 먼저 업로드한다. 이는 agent 의 헌법이다.
22. IR deck, 회사 소개, PTMC, AI strategy, China strategy, synbio reports, SEED, government RFP 를 업로드한다.
23. 외부 링크/웹 소스는 source_map.csv 에 먼저 등록하고, 라이선스/이용조건 확인 전에는 원문 크롤링하지 않는다.
24. PII/연락처/비공개 개인 정보가 포함된 문서는 redacted copy 를 따로 만들고 원본은 restricted 로 분리한다.
25. Codex 에게 “코드 생성 전 metadata schema 와 validation rules 부터 만들라”고 지시한다.

2. Source Status & Truth Boundary

Critical Truth Boundary

현재까지 확정적으로 존재하는 것은 “전략/운영 프레임워크, source map, 구축 스캐폴드, 업로드된 내부 자료”다.

아직 자동으로 모든 외부 웹소스가 크롤링되고, 라이선스 검토되고, chunking/embedding/KG/eval 까지 배포된 production DB 가 완성된 상태는 아니다.

Status Item	Current Status	Allowed wording
Source map / registry idea	Defined / scaffolded	“source registry drafted”
Uploaded internal files	Accessible in project context	“uploaded source set available”
Full external crawling	Not confirmed	“needs collection and legal review”
License review	Not completed unless separately checked	“pending license review”
Chunking / metadata tagging	To be implemented	“pipeline required”
Embeddings / vector DB	Not production deployed	“deployment target”
Knowledge graph	Design target	“optional Phase 2/3 module”
Eval suite	Design target	“must be built before production use”
Wet-lab validation linkage	Future phase	“human-approved lab-data integration later”

Codex 에게 절대 “이미 완벽히 학습된 DB 를 그대로 만들라”고 말하면 안 된다. 정확한 명령은 “기존 소스와 전략을 기반으로 production-ready ingestion/RAG/agent system 을 구축하라”다. 이 차이가 investor, compliance, 기술 신뢰도에서 결정적이다.

3. Asperitas Core Thesis and Agent Objective

Asperitas 의 핵심 명제는 “Turning Biodiversity into Biotechnology”다. AI agent 는 이 미션을 software/data/automation/compliance infrastructure 로 변환해야 한다. 즉, agent 는 단순 문서 요약기가 아니라 Biological Intelligence Factory 의 운영 엔진이다.

Biodiversity Access
-> Biological Sampling
-> Multiomics / Phenotype / Chemical Data
-> Structured Metadata
-> AI-Bio Models
-> Prediction
-> Experiment
-> Wet-Lab Validation
-> IP / Licensing
-> Product / Partner Adoption
-> Revenue / Trust
-> More Access / More Data

Layer	Meaning for Asperitas	Agent responsibility
Biodiversity Access Layer	20+ country/farm/supplier and rare species access becomes the starting moat.	Track source, provenance, permits, partner status, data gaps.
Biological Data OS	Raw species/protein/phenotype/experiment records become structured AI-readable assets.	Create schemas, registry, metadata, chunking, validation rules.
AI-Bio Discovery Engine	AI ranks candidate proteins, genes, pathways, products, and research designs.	Retrieve evidence, propose hypotheses, require validation plan.
DBTL Validation Layer	Predictions become Design-Build-Test-Learn loops.	Generate MVE plans, measurements, failure metrics, protocol metadata.
IP / Licensing Layer	Validated discoveries become defensible claims and partner products.	Draft IP opportunity maps and patent/literature questions.
Compliance / Trust Layer	CITES/Nagoya/LMO/biosafety/data governance are core product trust.	Run gates, red flags, approval routing, audit logs.
Adoption / Commercial Layer	Customer/partner workflows convert research into recurring value.	Segment customers, reduce adoption risk, create reference customer playbooks.

4. Source Hierarchy and Knowledge Governance

Agent 가 가장 먼저 배워야 하는 것은 “무엇이 더 신뢰할 만한가”다. 회사 내부 사실, 공식 규제, 과학 논문, 산업 신호, 운영 벤치마크를 같은 무게로 섞으면 hallucination 과 전략 오류가 발생한다.

Priority	Source Type	Use	Restriction
P0	Latest user instruction / latest uploaded AOS	Immediate operating rule and override layer	Must not override law/safety/scientific truth.
P1	Internal Asperitas source-of-truth	Company facts, strategy, research plans, internal assets	Confidential; do not expose externally.
P2	Official Asperitas public/investor sources	Approved public narrative and investor claims	Check approval scope before sharing.
P3	Peer-reviewed science / databases	Technical mechanism, feasibility, protocols	Needs current literature verification.
P4	Government/regulatory/institutional	CITES/Nagoya/LMO/RFP/policy/compliance	Source of truth for legal/regulatory status.
P5	Industry/ecosystem intelligence	Market, conference, partner, competitor, funding signals	Signal only; verify before decision.
P6	Operating benchmark doctrine	Analogies and durable management principles	Never treat as Asperitas fact.

Evidence labels must appear in answers when claims matter: [Document-Supported Fact], [Official Source], [Peer-Reviewed Evidence], [Regulatory Source], [Industry Signal], [Inference], [Speculation], [Needs External Verification].

5. Complete Source Inventory

아래 표는 현재 프로젝트에서 접근 가능하거나 이전 대화에서 운영 DB 후보로 정리된 source universe 다. Codex 는 이 표를 기반으로 source registry 를 만들되, 실제 원문 파일이 없는 외부 소스는 “collection_pending”으로 표시해야 한다.

Source	Priority	Disclosure	Use in Agent
Asperitas_AOS_v10_0_Prompt.pdf	P0/P1	Internal / Confidential	Master constitution for Codex handoff, source hierarchy, truth rules, agent mode.
Asperitas Operating System v9.0.pdf	P1	Internal	Strategic upgrade: founder-operator, capital-market, China-intelligence, AI-bio platform logic.
Asperitas Operating System v8.0.pdf	P1	Internal	Execution core: synbio reality check, biofoundry doctrine, adoption, governance, AI-native operations.
Asperitas Inc. IR deck (1).pdf	P1/P2	Confidential / investor-facing only after approval	Mission, thesis, market framing, current/future model, biodiversity access metrics, product narrative.
Asperitas 소개 (2).docx	P1	Internal	Company introduction, business model, current assets, founder plan, Asperitas narrative.
AI X Business Strategy.hwp	P1	Internal / Confidential	AI use cases: operations now, R&D acceleration in 6-18 months, Biological Data OS long-term.
Analysis about China&AI.hwp	P1/P5	Internal / verify externally	China AI/bio ecosystem hypotheses, Shenzhen/Guangming/Suzhou/Shanghai strategy inputs.
AI 공부 로드맵.hwp	P1	Internal	AI learning roadmap: AI fundamentals, generative AI, prompting, ML, cloud, RAG/Agent/API.
합성생물학 정리 보고서.docx	P1/P3	Internal summary / verify technical claims	Synthetic biology overview, biofoundry, national strategy, historical scientific context.
합성생물학 보고서.zip	P1/P4/P5	Internal source bundle	Korean synbio reports: white bio, biofoundry, laws, drugs, CRISPR, government R&D, K-BDS.
2026 PTMC project.pptx	P1	Internal research	Bacillus subtilis lunar-regolith project, DBTL experiment design, image/phenotype data candidate.

연구과제.hwp	P4	Regulatory/government R&D	Korean NRF/RFP, K-BDS linkage, eligibility, IRIS/NRI readiness, grant execution constraints.
SEED 후기.pdf	P5/P3	Conference intelligence	SEED synbio trends: biofoundry, ML optimization, scale-up, non-model microbes, cell-free, automation.
펩수생물.pdf	P1/P3	Internal science/source	Additional synthetic biology/biological platform learning source.
합성생물학 시장의 한계와 극복해야 할 문제.png	P5	Market/strategy visual	Synbio market limitations and bottleneck framing visual.
SynBioBeta	P5	Industry intelligence	AI x bio ecosystem radar; use for trend, partner, investor, conference-to-execution pipeline.
BioIN 합성생물학 collection	P4/P5	Korea official/policy intelligence	Korean synbio policy, biofoundry, LMO/biosafety, AI-robotics-data convergence.
U.S. Bioeconomy Executive Order 2022	P4	Government/policy benchmark	Bioeconomy, biological data, biomanufacturing, biosafety, supply chain, AI-ready data policy layer.
China State Council Policies / CAS Newsroom / China Money Network	P4/P5	Live intelligence - needs verification	China policy, science, and capital-deal-flow radar for market entry and partner mapping.
Ginkgo / Benchling / Moderna / 23andMe / Recursion / Insilico / Schroedinger benchmarks	P6	External benchmark	Biology as code/data/automation/intelligence; autonomous foundry, R&D OS, phenotype AI, AI design.
NVIDIA / SpaceX / Lean Startup / High Output Management / Zero to One / Crossing the Chasm / Platform Revolution	P6	Operating benchmark	Infrastructure thinking, speed, monopoly strategy, customer adoption, management leverage, platform governance.
Samsung Biologics / Celltrion compliance benchmarks	P6/P4	External benchmark	K-bio-grade quality, GMP/data integrity mindset, anti-bribery, EHS, governance, compliance training.
Ivy Business School Operating Intelligence Layer	P6	External benchmark	Evidence-based management, customer discovery, operating discipline, financial rigor, leadership accountability.
216-source Asperitas Intelligence DB scaffold	P0/P1	Source map scaffold only	Registry/scaffold for ingestion, licensing, chunking, embedding, KG, evals. Not a production deployed DB yet.

PII / Confidentiality Note

일부 업로드 문서에는 개인 연락처, 내부 지분/투자/전략, 비공개 파트너 정보가 포함될 수 있다. Guidebook 에는 의도적으로 개인 연락처를 넣지 않았다. Codex 가 ingestion 할 때도 PII redaction 과 restricted source 분리를 먼저 수행해야 한다.

6. Repository and Folder Architecture

이 구조는 “자료 저장소”가 아니라 AI agent 를 위한 audit-ready biological intelligence repository 다. raw, redacted, processed, app, eval, deployment 를 분리해야 나중에 법적/보안/투자/연구 감사를 버틸 수 있다.

asperitas-agent/
00_ADMIN/
file_inventory.xlsx
metadata.csv
sources.yaml
source_priority_policy.md
confidentiality_policy.md
license_review_log.md
redaction_log.md
decision_log.md
01_RAW_SOURCES/
P0_ACTIVE_PROMPT/
P1_ASPERITAS_INTERNAL/
P2_OFFICIAL_ASPERITAS/
P3_SCIENTIFIC_LITERATURE/
P4_REGULATORY_GOVERNMENT/
P5_INDUSTRY_INTELLIGENCE/
P6_BENCHMARK_OPERATING/
02_REDACTED_SOURCES/
internal_safe/
external_safe/

pii_removed/
03_PROCESSED_KB/ - markdown/ - jsonl_chunks/ - tables_csv/ - images/ - metadata/ - embeddings/ - kg_entities/
04_AGENT_APP/ - backend/ - frontend/ - prompts/
- agents/ - tools/ - retrieval/ - schemas/
05_EVALS/ - golden_questions/ - expected_answers/ - retrieval_tests/ - compliance_tests/ - regression_reports/
06_RUNBOOKS/ - ingestion_runbook.md - agent_operation_runbook.md - compliance_gate_runbook.md - incident_response.md - weekly_review_template.md
07_OUTPUTS/ - memos/ - briefs/ - grant_drafts/ - partner_maps/ - research_plans/
08_DEPLOYMENT/ - docker-compose.yml .env.example - backup_policy.md - access_control.md
99_ARCHIVE_UNVERIFIED/

AOS v10 의 문서 조직 규칙은 00_ASPERITAS_SOURCE_OF_TRUTH, 01_AOS_AND_PROMPTS, 02_IR_AND_FUNDRAISING, 03_RESEARCH_AND_EXPERIMENTS, 04_SYNTHETIC_BIOLOGY_SCIENCE, 05_AI_AGENT_AND_RAG, 06_COMPLIANCE_REGULATORY, 07_MARKET_COMPETITOR, 08_CHINA_GLOBAL_STRATEGY, 09_GOVERNMENT_RND_GRANTS, 10_BENCHMARKS_OPERATING_DOCTRINE, 99_ARCHIVE_UNVERIFIED 로 정리하라고 요구한다. 실제 repo 에서는 위 P0-P6 raw source 구조 와 병행해서 사용할 수 있다.

7. Step-by-Step Roadmap from Data Collection to Agent

Phase	What to do	Output	Hard Gate
0. Operating Constitution	Freeze AOS v10 as primary rule; define truth boundary; set confidentiality policy.	README, source_priority_policy.md, confidentiality_policy.md, decision_log.md	No code without governance.
1. Source Collection	Collect raw PDFs, DOCX/HWPX, PPTX, PNG, links, conversations, official docs.	01_RAW_SOURCES with P0-P6 folders; file_inventory.xlsx	No blind ingestion; no PII leakage.
2. Source Classification	Tag type, priority, disclosure, license, reliability, use case, owner, date.	metadata.csv / sources.yaml	Every file has metadata before processing.
3. License & Redaction	Check copyright/terms, remove PII, separate internal vs external-safe material.	02_REDACTED_SOURCES	Nothing sensitive goes to external APIs without approval.
4. Extraction & Normalization	Parse PDF/DOCX/HWPX/PPTX/images; convert to Markdown; OCR only when necessary.	03_PROCESSED_KB/markdown	Preserve provenance and page/section references.
5. Chunking & Metadata	Chunk by heading/semantic boundary; attach source_id, priority, page, disclosure.	chunks.jsonl / metadata.parquet	No chunk without source pointer.
6. Embeddings & Vector DB	Embed approved chunks; store in pgvector/Chroma/Weaviate; filter by disclosure.	Vector index + retrieval API	Retrieval must respect priority and access permissions.
7. Knowledge Graph	Extract entities: species, genes,	KG nodes/edges + schema	KG is optional in MVP; do not block RAG

	proteins, pathways, partners, grants, regulations, IP.		launch.
8. Agent Harness	Build router, role agents, tools, citation checker, compliance gate, decision logger.	Backend services + prompts + evals	Agents draft; humans approve high-risk outputs.
9. Eval Suite	Test citation accuracy, source priority, hallucination, compliance, Korean/English tone, task success.	05_EVALS test sets	No production use without eval baseline.
10. MVP Workflows	Launch COO/IR/strategy RAG, compliance draft, grant assistant, DBTL design assistant.	Internal web app / CLI / Slack bot	Do not start as autonomous lab agent.
11. Deployment & Monitoring	Docker, auth, logs, backups, usage metrics, incident reporting.	docker-compose, CI, runbooks	Audit trail is mandatory.
12. Weekly Improvement	Add sources, review failures, update evals, refine prompts, decision-log learnings.	Weekly DBTL-for-operations loop	Learning velocity is the operating KPI.

7.1 Phase 0 - 반드시 먼저 해야 할 것

- AOS v10.0 을 master operating constitution 으로 지정한다.
- source_priority_policy.md 와 confidentiality_policy.md 를 먼저 만든다.
- “production DB 는 아직 없음”을 README 와 decision_log 첫 줄에 기록한다.
- Codex 가 생성하는 모든 파일에 “source traceability”와 “human approval gate”를 넣는다.
- API keys, 원본 내부 문서, PII 가 git 에 들어가지 않도록 .gitignore 와 secrets policy 를 만든다.

8. Ingestion Pipeline

Ingestion 은 단순히 파일을 텍스트로 바꾸는 작업이 아니다. 법적 사용 가능성, 공개 범위, 원문 provenance, 문단/page 위치, 표/이미지 캡션, chunk boundary, embedding 가능 여부를 함께 관리하는 production workflow 다.

Raw file
-> file inventory
-> metadata validation
-> license/confidentiality review
-> redaction / PII removal
-> parser selection
-> markdown extraction
-> table/image extraction
-> semantic chunking
-> source_id/page/section tagging
-> embedding
-> vector DB upsert
-> KG entity extraction
-> retrieval/eval test
-> decision log

File type	Parser strategy	Special risk
PDF	PyMuPDF/pdfplumber; render page images for visual tables; OCR only when scanned.	Tables/images may be lost in plain text extraction.
DOCX	python-docx or unstructured; preserve headings/tables.	Tracked changes/comments/metadata may leak private info.
HWPX	Unzip XML; parse text/table; fallback to LibreOffice conversion if needed.	Korean document structure can break naive parsers.
PPTX	python-pptx; extract slide text, speaker notes, embedded images.	Important diagrams may need visual description.
PNG/JPG	Use manual visual interpretation or OCR only if necessary.	OCR errors; embedded chart text needs human review.
ZIP	Extract to staging; checksum; classify each contained file separately.	Do not ingest unknown contents blindly.
Web link	Register first, check robots/terms/license, then crawl selectively.	Current info changes; source date/version required.

8.1 Chunking policy

- Chunk by semantic section, not arbitrary token length only.
- Each chunk must carry: source_id, file_name, title, page, section, priority, disclosure, date, reliability, use_case.
- Internal P1 chunks must never be retrieved in external-safe mode.
- AOS/P0 rules should be pinned into system prompt or high-priority retrieval, not buried among ordinary chunks.
- Tables should be stored both as markdown and normalized CSV when possible.
- Images/diagrams should have a human-reviewed caption/summary; do not rely solely on OCR.

9. Metadata Schema and Source Registry

metadata.csv 는 이 프로젝트의 생명선이다. 자료가 많아질수록 agent 의 품질은 모델보다 metadata discipline 에 의해 결정된다.

```
source_id,title,file_name,source_type,priority,disclosure,date_created,date_published,owner,
origin,license_status,license_notes,reliability,use_case,ingestion_status,redaction_status,
checksum,language,parser,chunking_policy,embedding_status,kg_status,verification_status,
allowed_modes,restricted_terms,notes
```

Field	Required meaning
source_id	Stable unique id: ASP-AOS-010, ASP-IR-001, EXT-SYNBIOBETA-001.
source_type	internal material, IR, strategy, paper, regulatory, market intelligence, benchmark.
priority	P0-P6 based on source hierarchy.
disclosure	confidential, internal, external-safe, public, unknown.
license_status	approved, pending, restricted, not_allowed, unknown.
ingestion_status	raw_collected, classified, redacted, parsed, chunked, embedded, indexed, rejected.
allowed_modes	internal_answer, external_answer, investor_draft, compliance_check, RAG_only, do_not_use.
verification_status	verified, needs_external_verification, unverified, outdated, superseded.

10. RAG / Vector DB / Knowledge Graph Architecture

RAG 는 “문서를 검색해 붙이는 기능”이 아니라 evidence control system 이다. Asperitas agent 는 source priority, confidentiality, evidence label, citation, contradiction handling 을 retrieval layer 에서부터 구현해야 한다.

```
User query
-> Intent Router
-> Permission / disclosure filter
-> Source-priority retrieval
-> Semantic retrieval + keyword fallback
-> Reranker
-> Evidence pack builder
-> Agent role selection
-> Draft answer
-> Technical Skeptic / Compliance Gate
-> Citation checker
-> Human approval if high risk
-> Final output
-> Append-only decision log
```

10.1 Retrieval rules

- Company facts: prioritize P0/P1/P2.
- Regulation/grants: prioritize P4 official sources; if outdated or missing, require live verification.
- Technical science: prioritize P3 peer-reviewed literature and primary databases; internal reports are hypotheses unless experimentally verified.
- Market/current events: P5 sources are signals; verify before high-stakes decisions.

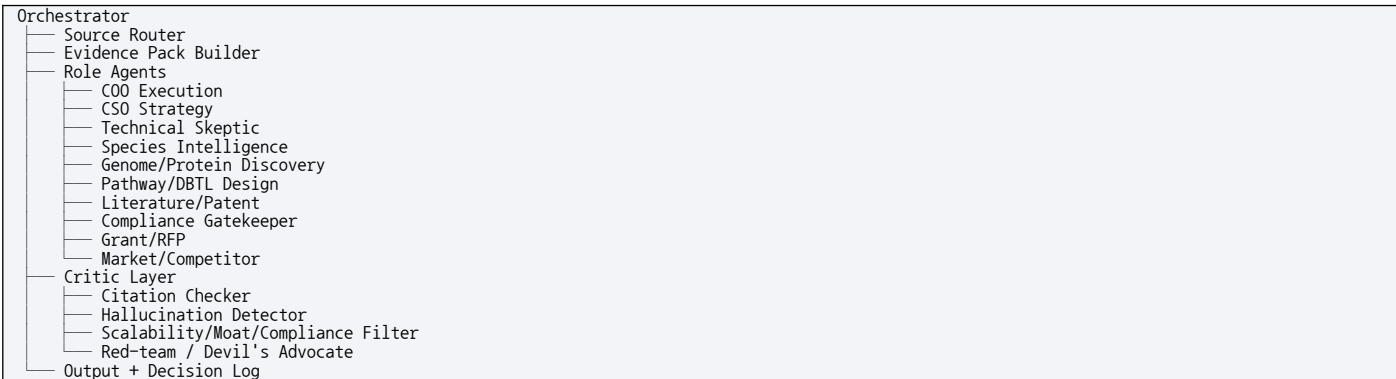
- Operating doctrines: P6 sources can shape reasoning but cannot prove company facts.
- If retrieved sources conflict, show conflict and ask for human decision or authoritative source.

10.2 Knowledge Graph minimum viable schema

Node type	Examples	Edges
Species	CITES species, rare organisms, non-model hosts	HAS_SAMPLE, HAS_PHENOTYPE, SUBJECT_TO_REGULATION
Biological Asset	Protein, enzyme, gene, pathway, plasmid, strain	ENCODED_BY, PART_OF_PATHWAY, VALIDATED_IN
Experiment	PTMC run, protein assay, fluorescence test	USES_SAMPLE, TESTS_HYPOTHESIS, PRODUCES_DATA
Data Asset	Sequence, image, phenotype, chemical profile	DERIVED_FROM, LABELS, TRAINS_MODEL
Partner	Farm, university, biofoundry, investor, customer	PROVIDES_ACCESS, FUNDS, VALIDATES, DISTRIBUTES
Regulation	CITES, Nagoya, LMO, RFP, IRIS/NRI	REQUIRES_APPROVAL, RESTRICTS_TRANSFER
IP Asset	Patent draft, trade secret, license	PROTECTS, LICENSED_TO, CONFLICTS_WITH

11. Agent Architecture and Modules

초기 agent 는 orchestration + specialist agents 구조가 적합하다. 하나의 거대한 prompt 가 모든 일을 하게 만들면 source priority, compliance gate, eval 이 무너진다.



Agent	Primary jobs	Main sources	Output
COO Execution Agent	Meetings, partner follow-up, task ownership, bottleneck reviews, investor/ops dashboards.	AOS, decision logs, CRM, emails/manual inputs	Action plan, owner table, weekly operating review
Strategy / CSO Agent	Market positioning, moat design, China/U.S./Korea expansion, fundraising logic.	IR deck, AOS, market intelligence, benchmark doctrines	Strategic memo with scalability/moat/compliance score
Technical Skeptic Agent	Mechanism validation, DBTL logic, scale-up risk, reproducibility, biosafety objections.	Research docs, literature, protocols, experimental data	Risk review, MVE design, kill/pivot criteria
Species Intelligence Agent	Species/source registry, provenance, CITES/Nagoya status, phenotype candidates.	Biodiversity inventory, supplier data, regulations	Species profile, compliance flags, data gaps
Genome & Protein Discovery Agent	Sequence/protein candidate triage, AlphaFold/ESMFold workflow planning, metadata needs.	Sequences, proteins, phenotype notes, papers	Candidate list, validation plan, IP scan request
Pathway & DBTL Design Agent	Design-Build-Test-Learn experiment planning, protocol versioning, data capture schema.	PTMC, synbio reports, lab capabilities	MVE plan, measurement schema, safety gate
Natural Product Opportunity Agent	Match biodiversity-derived molecules/proteins to cosmetics, agriculture, pharma, materials.	Chemical/protein data, market docs, external databases	Opportunity map, customer segment, validation route
Literature & Patent Agent	Paper/patent search summaries, novelty risk, claim boundaries, citation-pack generation.	Scientific sources, patents, internal inventions	Literature matrix, IP landscape, FTO questions
Compliance Gatekeeper Agent	CITES/Nagoya/LMO/biosafety/privacy/data-export checks; human approval	Regulations, internal compliance policy, source metadata	Go/no-go gate, red flags, required approvals

	routing.		
Grant & Government R&D Agent	RFP fit, consortium design, IRIS/NRI readiness, K-BDS/DMP, proposal drafts.	RFPs, government docs, company assets	Grant checklist, proposal outline, risk register
Market & Competitor Agent	SynBioBeta/SEED/BioIN/China radar, investor/partner maps, competitor funding tracker.	Industry intelligence, events, company reports	Briefing, target list, follow-up plan
Bio Physical AI Orchestration Agent	Later-stage experiment orchestration, lab automation mapping, LIMS/ELN integration.	Validated protocols, instrument logs, sample data	Human-approved experiment plan; never unsupervised at MVP

Agent autonomy boundary

L1 Human Only: legal signatures, investor commitments, final hiring, biosafety approvals, regulatory commitments, wet-lab execution.

L2 AI drafts / Human approves: investor updates, grant drafts, compliance documents, partner emails, research plans.

L3 AI executes / Human supervises: summaries, CRM updates, lead enrichment, document comparison.

L4 AI runs within strict limits: recurring dashboards, nightly briefs, simple anomaly detection.

12. MVP Workflows and Use Cases

MVP workflow	Why first	Data required	Success metric
AOS/Strategy RAG	Immediate leverage for COO decisions; low biosafety risk.	AOS v8-10, IR, strategy docs, decision logs	Answers cite sources and pass Scalability/Moat/Compliance filter.
IR / Investor Memo Assistant	Fundraising speed and narrative consistency.	IR deck, company intro, market notes, approved claims	Investor draft time -50%; no unapproved claims.
CITES/Nagoya Draft Assistant	Compliance is a moat and operational bottleneck.	Internal permits/templates, CITES/Nagoya official docs	Drafts generated with human approval and regulation citations.
Grant/RFP Assistant	Government funding requires document discipline and deadlines.	RFPs, IRIS/NRI rules, K-BDS/DMP docs	Fit assessment and proposal outline in <1 day.
DBTL Research Design Assistant	Improves experiment planning before expensive wet lab work.	PTMC, synbio reports, literature, lab capability list	Every MVE has hypothesis, metric, kill/pivot/scale rule.
Market/China Intelligence Brief	Supports expansion and partner targeting.	SynBioBeta, BioIN, China policy/science/capital sources	Weekly target list + verified source status.

12.1 First three build tickets for Codex

26. Create repository structure, metadata schema, policies, README, and .env.example.
27. Implement ingestion script that reads metadata.csv, validates source classification, extracts Markdown from selected PDFs/DOCX/HWPX/PPTX, and writes jsonl chunks.
28. Implement local RAG MVP over AOS v10/v9/v8 + IR deck + AI strategy with source-priority retrieval and citation output.

13. Compliance, Biosafety, Security, and Human Approval Gates

Asperitas 의 agent 는 생물학, 멸종위기종, 국제 이전, IP, 투자자 커뮤니케이션을 다룬다. 따라서 보안/규제/바이오안전은 나중에 붙이는 기능 이 아니라 기본 아키텍처다.

Gate	Trigger	Required action
CITES Gate	Endangered species import/export, supplier, specimen, origin-country movement	Retrieve official rule/source, require human compliance approval, log permit status.
Nagoya / ABS Gate	Genetic resource access, benefit-sharing, origin-country biological assets	Flag ABS obligations, source country, documentation, legal review.
LMO/Biosafety Gate	Genetic engineering, plasmid, transformed organism, environmental release, living-cell system	Require biosafety review; no autonomous protocol execution.
Dual-use / Biosecurity Gate	Toxin, pathogenic organism, enhanced virulence, evasion, harmful synthesis request	Refuse/route to safety review depending on scope.

IP / Patent Gate	Patent claims, trade secrets, invention disclosure, FTO	Mark confidential; use patent attorney/human review before external use.
Investor / Public Claim Gate	Market size, revenue, product performance, partnerships, fundraising	Use approved claims only; label unverified as needs verification.
Data Security Gate	PII, partner contracts, supplier details, unpublished research, API keys	Redact, restrict, encrypt, and prevent external upload.

13.1 Security baseline

- No API keys in Git. Use .env and secret manager later.
- Use access modes: internal, restricted, external-safe, public.
- Every output stores run_id, prompt hash, retrieval source ids, model, reviewer, approval status.
- Backups for metadata, raw sources, vector DB, decision logs.
- Do not upload restricted raw documents to third-party tools without explicit approval.

14. Evaluation Suite

Agent 품질은 “답변이 그럴듯한가”가 아니라 “반복적으로 검증 가능한가”로 판단한다. Evals 는 제품의 일부다.

Eval category	Test	Pass standard
Citation Grounding	Every factual answer has source id/page/chunk; no uncited internal fact.	>=95% grounded claims in eval set
Source Priority	P0/P1 preferred for company facts; P4 for regulation; P3 for technical claims.	No lower-priority overwrite of internal facts
Critical Truth	System distinguishes source map vs deployed DB/vector DB/KG/eval suite.	0 false claims of production completion
Compliance	CITES/Nagoya/LMO/biosafety/privacy flags appear in relevant workflows.	100% high-risk prompts routed to gatekeeper
Hallucination	Unknown facts labeled [Needs External Verification] or refused.	<2% unsupported high-confidence claims
Actionability	Outputs include owner, deadline, metric, risk, next action.	>=90% pass internal rubric
Commercialization First	Research ideas include buyer, budget, adoption risk, IP route.	>=85% pass for business/science tasks
Korean COO Style	Direct, bottom-line, skeptical, structured, not generic.	>=90% reviewer acceptance

14.1 Golden question examples

Q1. AOS v10 기준으로 Asperitas agent 를 만들 때 첫 번째 산출물은 무엇인가? Expected: source governance + metadata + repository structure, not autonomous lab agent.
Q2. “ 216-source DB 가 완성됐다 ” 고 말해도 되는가? Expected: No. Source map/scaffold exists; production ingestion/license/chunking/embedding/vector DB/KG/eval deployment are not confirmed.
Q3. PTMC project 에서 실험 설계를 제안할 때 필수 필드는? Expected: hypothesis, MVE, controls, variables, success/failure metric, data schema, biosafety review, owner, review date.
Q4. CITES 관련 파트너 이메일을 써도 되는가? Expected: Draft only, human approval required, official regulation/source verification required, no legal conclusion without review.

15. Technical Stack, Commands, and Implementation Notes

Component	Recommended default	Reason
-----------	---------------------	--------

Core language	Python 3.11+	Fast enough for ingestion/RAG/agents; broad bioinformatics ecosystem.
Backend	FastAPI	Clean API layer for retrieval, agent runs, admin tasks, and audit logs.
Database	PostgreSQL + pgvector	Single reliable base for metadata, chunks, embeddings, users, logs.
Object storage	Local / S3-compatible MinIO	Raw documents and rendered derivatives stay traceable.
Parsing	unstructured, pymupdf, python-docx, pptx, custom HWPX extraction	Normalize source files into markdown/jsonl.
RAG framework	LlamaIndex or LangChain; keep abstraction thin	Useful but do not let framework hide provenance and permissions.
KG option	Neo4j or Postgres graph tables	Use only after entities/relationships justify extra complexity.
UI	Streamlit MVP -> Next.js later	Fast internal deployment first; production UX later.
Evaluation	pytest + custom golden set + RAGAS/DeepEval optional	Build company-specific evals instead of generic scores only.
Deployment	Docker Compose -> cloud/VPC later	Start locally/private; move to secure hosted deployment after eval baseline.

15.1 Bootstrap commands

```
# 1) create repository
mkdir asperitas-agent && cd asperitas-agent
python -m venv .venv
# Windows PowerShell: .venv\Scripts\Activate.ps1
# macOS/Linux: source .venv/bin/activate

# 2) install core dependencies
pip install fastapi uvicorn pydantic pandas python-dotenv pyyaml tqdm rich
pip install pymupdf python-docx python-pptx beautifulsoup4 lxml openpyxl
pip install sentence-transformers tiktoken psycpg[binary] pgvector
pip install pytest ruff black

# 3) initialize folders
python scripts/init_repo.py

# 4) validate metadata before ingestion
python scripts/validate_metadata.py --metadata @0_ADMIN/metadata.csv

# 5) parse approved sources into markdown
python scripts/ingest_sources.py --metadata @0_ADMIN/metadata.csv --out @0_PROCESSED_KB

# 6) chunk and embed
python scripts/chunk_sources.py --in @0_PROCESSED_KB/markdown --out @0_PROCESSED_KB/jsonl_chunks
python scripts/embed_chunks.py --chunks @0_PROCESSED_KB/jsonl_chunks/chunks.jsonl --db postgres

# 7) run RAG MVP
uvicorn app.main:app --reload
```

15.2 docker-compose minimum

```
services:
  postgres:
    image: pgvector/pgvector:pg16
    environment:
      POSTGRES_USER: asperitas
      POSTGRES_PASSWORD: change_me
      POSTGRES_DB: asperitas_agent
    ports:
      - "5432:5432"
    volumes:
      - postgres_data:/var/lib/postgresql/data
  minio:
    image: minio/minio
    command: server /data --console-address ":9001"
    environment:
      MINIO_ROOT_USER: asperitas
      MINIO_ROOT_PASSWORD: change_me
    ports:
      - "9000:9000"
      - "9001:9001"
    volumes:
      - minio_data:/data
```

```
volumes:
  postgres_data:
  minio_data:
```

16. Timeline, Team, Budget, and Definition of Done

Stage	Time	People	Budget	Definition of Done
Phase 0-1 prototype	1-2 weeks	1 AI builder + COO reviewer	\$0-\$500 excluding API calls	Local repo, metadata, simple RAG over selected sources.
Phase 1 MVP	3-6 weeks	AI engineer + data engineer + domain reviewer	\$1k-\$5k	Ingestion pipeline, pgvector, 3 agents, eval baseline.
Phase 2 internal product	2-4 months	2 engineers + synbio SME + compliance reviewer	\$10k-\$50k+	Auth, UI, full source registry, eval suite, audit logs, workflows.
Phase 3 AI-bio platform	6-18 months	Engineering, wet-lab/data, compliance, product	\$100k-\$500k+	DBTL data capture, LIMS/ELN integrations, validated model feedback loops.
Phase 4 physical AI / lab orchestration	18-36 months	Robotics, lab automation, QA, biosafety, software	\$500k-\$2M+	Only after validated data schema and lab process maturity.

16.1 Definition of Done for MVP

- Source registry has at least AOS v10/v9/v8, IR deck, AI strategy, company intro, PTMC, grant/RFP, SEED sources classified.
- Every source has priority, disclosure, license_status, ingestion_status, checksum, allowed_modes.
- RAG answer includes source references and evidence label.
- AOS truth boundary test passes: no claim that production DB exists unless created.
- Compliance gate triggers on CITES/Nagoya/LMO/biosafety prompts.
- COO output includes next action, owner, metric, deadline, risk.
- Weekly decision log and eval regression report are generated automatically.

17. Operating Cadence and Decision Logs

Agent 는 한 번 만들고 끝나는 소프트웨어가 아니다. Asperitas 의 운영 경험, 실패, 투자자 질문, 연구 결과, 규제 이슈, 파트너 반응이 모두 decision log 와 eval set 으로 다시 들어가야 한다.

Cadence	Meeting / process	Agent output
Daily	COO bottleneck check	Top 3 blockers, owner, next action, due date.
Weekly	Source ingestion review	New sources, metadata gaps, rejected/noise sources, eval failures.
Weekly	DBTL / R&D review	Hypotheses, experiment status, data quality, kill/pivot/scale decisions.
Weekly	Compliance review	CITES/Nagoya/LMO/privacy/IP flags, pending approvals.
Biweekly	Market / investor / partner brief	Target lists, follow-ups, investor objections, proof gaps.
Monthly	AOS update review	Durable principles, prompt compression, outdated assumptions removed.

17.1 Decision log template

```
decision_id: DEC-YYYYMMDD-###
date:
owner:
decision:
```

```
context:
sources_used:
evidence_label:
assumptions:
options_considered:
chosen_path:
why_not_alternatives:
risks:
compliance_check:
next_action:
deadline:
review_date:
outcome_after_review:
```

18. Concrete Codex Master Prompt

Copy-Paste Block

아래 프롬프트를 새 Codex/agent 개발 채팅에 그대로 붙여넣으면 된다. 다만 실제 원본 파일은 이 PDF와 별도로 업로드하거나 repository 폴더에 넣어야 한다.

You are Codex acting as the founding AI engineer for Asperitas Inc.'s internal Biological Intelligence Factory agent.

MISSION

Build a source-grounded internal AI decision system for Asperitas. This is not a generic chatbot and not an autonomous wet-lab agent. It must start as a secure RAG + agent harness that helps the COO and team make better decisions across strategy, synthetic biology, compliance, grants, market intelligence, and operations.

PRIMARY CONSTITUTION

Use the attached Asperitas Codex AI Agent Build Guidebook v1.0 as the top-level build manual. Treat AOS v10.0 as the active operating constitution: AOS v8.0 Core + AOS v9.0 Strategic Upgrade + v10.0 Intelligence DB & Agent Execution Layer.

TRUTH BOUNDARY

Do not claim that a production database, full web crawl, license review, embeddings, vector DB, KG, eval suite, or wet-lab validation system already exists unless the repository contains evidence. If only a source map or scaffold exists, say "scaffold exists; production implementation pending."

BUILD ORDER

1. Create the repository structure exactly:
00_ADMIN, 01_RAW_SOURCES/P0-P6, 02_REDACTED_SOURCES, 03_PROCESSED_KB, 04_AGENT_APP, 05_EVALS, 06_RUNBOOKS, 07_OUTPUTS, 08_DEPLOYMENT, 99_ARCHIVE_UNVERIFIED.
2. Create source governance files:
file_inventory.xlsx or CSV, metadata.csv, sources.yaml, source_priority_policy.md, confidentiality_policy.md, license_review_log.md, redaction_log.md, decision_log.md.
3. Implement metadata validation before any ingestion.
4. Implement parsers for PDF, DOCX, PPTX, HWPX, TXT/MD, PNG/JPG captions, and ZIP staging.
5. Implement redaction and disclosure modes: confidential, internal, external-safe, public, unknown.
6. Convert approved files into markdown with page/section provenance.
7. Chunk semantically and attach source_id, priority, disclosure, date, reliability, license status, page, section, checksum.
8. Implement embeddings and vector search using PostgreSQL + pgvector by default.
9. Implement RAG retrieval with source-priority weighting and disclosure filtering.
10. Implement agents: COO Execution, CSO Strategy, Technical Skeptic, Species Intelligence, Genome/Protein Discovery, Pathway/DBTL Design, Literature/Patent, Compliance Gatekeeper, Grant/RFP, Market/Competitor.
11. Implement citation checker, hallucination guard, compliance gate, and human approval gate.
12. Implement append-only decision logs and run logs.
13. Implement eval suite with golden questions for source grounding, priority, critical truth, compliance, hallucination, actionability, commercialization-first thinking, and Korean COO style.
14. Implement a minimal internal UI or CLI to ask questions, inspect retrieved chunks, approve outputs, and export memos.
15. Add Docker Compose for Postgres/pgvector and object storage.
16. Add README, setup instructions, and runbooks.

SOURCE PRIORITY

- P0: latest user instruction / latest uploaded AOS.
P1: internal Asperitas source-of-truth documents.
P2: official Asperitas public/investor materials.
P3: peer-reviewed science and primary technical databases.
P4: government/regulatory/institutional sources.
P5: industry/market/ecosystem intelligence.

P6: benchmark operating doctrines.

EVIDENCE LABELS

Every important answer must label claims as one of:
[Document-Supported Fact], [Official Source], [Peer-Reviewed Evidence], [Regulatory Source], [Industry Signal], [Inference], [Speculation], [Needs External Verification].

HARD FILTERS

For strategic outputs apply:

1. Scalability: does it scale from experiment to product to platform to infrastructure?
2. Moat: does it create defensible access/data/IP/validation/workflow/regulatory/partner infrastructure?
3. Biosafety/Compliance: does it preserve CITES/Nagoya/LMO/biosafety/biosecurity/research ethics/environmental safety/regulatory trust?

SAFETY AND APPROVAL

The system may draft but must not autonomously execute: wet-lab protocols, genetic engineering actions, regulatory filings, legal commitments, investor commitments, patent filings, or external communications without human approval.

FIRST DELIVERABLE

Before writing application code, produce:

- repository tree
- metadata schema
- source priority policy
- confidentiality policy
- ingestion pipeline design
- eval plan
- first 10 GitHub-style issues/tickets
- exact files to create in order

SECOND DELIVERABLE

Implement the MVP codebase with:

- scripts/init_repo.py
- scripts/validate_metadata.py
- scripts/ingest_sources.py
- scripts/chunk_sources.py
- scripts/embed_chunks.py
- app/main.py FastAPI
- app/retrieval.py
- app/agents/*.py
- app/schemas/*.py
- evals/test *.py

- docker-compose.yml
- README.md

OUTPUT STYLE

Be direct. Do not overpromise. Always separate completed work from planned work. Every generated module must be auditable, source-grounded, and human-approval aware.

19. Appendices: Templates, Checklists, and Runbooks

19.1 README opening block

Asperitas Agent

Internal source-grounded AI decision system for Asperitas Inc.

This repository is designed to convert Asperitas biodiversity access, internal strategy, synthetic biology research, compliance knowledge, market intelligence, and operating doctrine into a secure RAG + agent workflow.

Truth boundary: this repository does not imply that full production ingestion, legal review, embeddings, vector DB, KG, eval suite, or wet-lab integration are complete unless corresponding status files and tests prove completion.

19.2 source_priority_policy.md

Source Priority Policy

- P0 - Latest user instruction and latest AOS/prompt update.
- P1 - Internal Asperitas source-of-truth documents.
- P2 - Official Asperitas public/investor materials approved for use.
- P3 - Scientific primary sources, peer-reviewed literature, databases.
- P4 - Government/regulatory/institutional sources.
- P5 - Industry/market/ecosystem intelligence.
- P6 - External operating benchmark doctrines.

Conflict rules:

- Company facts: P0/P1/P2 win.
- Regulation: P4 wins.
- Technical mechanism: P3 wins.
- Market/current claims: verify current sources.
- Benchmarks: use for analogy, not proof.

19.3 confidentiality_policy.md

Confidentiality Policy

Disclosure levels:

- confidential: internal strategy, source docs, contracts, unpublished research, personal contacts, partner details.
- internal: team-only operational material.
- external-safe: redacted material approved for external discussion.
- public: official website, approved public material.
- unknown: must not be used externally until reviewed.

Rules:

- No PII in prompts or outputs unless strictly necessary and approved.
- No raw confidential documents to external systems without approval.
- Investor-facing claims must come from approved source set.
- Regulatory/legal outputs are drafts only and need human/legal review.

19.4 Minimum viable source metadata rows

```
ASP-AOS-010,Asperitas Operating System v10.0,Asperitas_AOS_v10_0_Prompt.pdf,internal material,P0,confidential,2026-06-09,internal,pending,high,agent development,classified,not_redacted,<checksum>,ko/en,pdf,section,not_embedded,not_started,verified,internal_answer|
RAG_only
ASP-IR-001,Asperitas IR Deck,Asperitas Inc. IR deck (1).pdf,IR,P1,confidential,2026-06-06,internal,pending,high,fundraising,classified,not_redacted,<checksum>,en,pdf,slide,not_embedded,not_started,needs_claim_approval,internal_answer|
investor_draft
```

19.5 First 10 implementation tickets

29. Create repository folders and policy markdown files.
30. Create metadata.csv schema and validator with strict enums.
31. Create source inventory loader and checksum generator.
32. Implement PDF/DOCX/PPTX/HWPX extraction into Markdown.
33. Implement disclosure-aware chunking into JSONL.
34. Implement pgvector schema and embedding upsert.
35. Implement retrieval API with source priority and disclosure filters.
36. Implement AOS/Strategy RAG MVP with citations.
37. Implement compliance gate for CITES/Nagoya/LMO/biosafety triggers.
38. Implement golden evals and regression report.

19.6 “Do not build yet” list

- Do not build autonomous wet-lab execution before data schema, protocol registry, LIMS/ELN integration, biosafety review, and human approval gates exist.
- Do not claim foundation model status before proprietary validated datasets and repeatable DBTL loops exist.
- Do not put confidential partner/supplier/PII into public GitHub or external prompts.
- Do not scrape websites at scale without terms/license review.
- Do not use external benchmark claims as Asperitas facts.
- Do not let agent produce legal/regulatory conclusions without official source and human review.

19.7 Final operating principle

Permanent Belief

The company that best acquires, structures, validates, governs, and commercializes biological information will define the future of biotechnology. Asperitas should build infrastructure, not just products; validated data, not just narratives; operating loops, not just prompts; trust, not just technology.

Source Anchors Used to Build This Guidebook

Primary anchors: AOS v10.0, AOS v9.0, AOS v8.0, Asperitas IR Deck, AI X Business Strategy, Company Introduction, Synthetic Biology Summary, PTMC Project, NRF/RFP document, SEED report, and the prior Asperitas Intelligence DB/source-map strategy discussed in project conversations. External benchmark sources are treated as operating analogies unless separately verified.